



UNIVERSITY OF WEST ATTICA

DEPARTMENT OF INFORMATION AND
COMPUTER ENGINEERING

DIGITAL DESIGN

WORKSHOP 1

LOGICAL GATES

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CHAPTER 1

A FEW WORDS ABOUT THE WORK

The work is based on the logic gates and their operation. More specifically, it is based on the AND , OR , NOT , NOR , NAND , XOR , XNOR logic gates, where the truth tables and the logic equation for each gate are presented. For the XNOR gate is tested using a 10 k Ω pull up resistor from the output to the 5V supply. Finally, propagation delay measurements are studied by connecting 4 AND gates in series, where a square pulse of frequency 100 kHz is applied to the input of the first gate.

CHAPTER 2

BIBLIOGRAPHY

- Design and implementation of logic circuits P
. Giannakopoulos L. Aslanoglou

CHAPTER 3

DESCRIPTION OF THE IMPLEMENTATION OF THE WORK

For the implementation of the project the simulator " multisim " was used for the design of the circuits. In more detail, cables, grounds, VCC sources, lamps, logic gates (AND , OR , NOR , NAND , XOR , XNOR , NOT), oscilloscope were used.

CHAPTER 4

EXERCISES

EXERCISE 1.2.1 GATE AND

THEORETICAL RESULTS

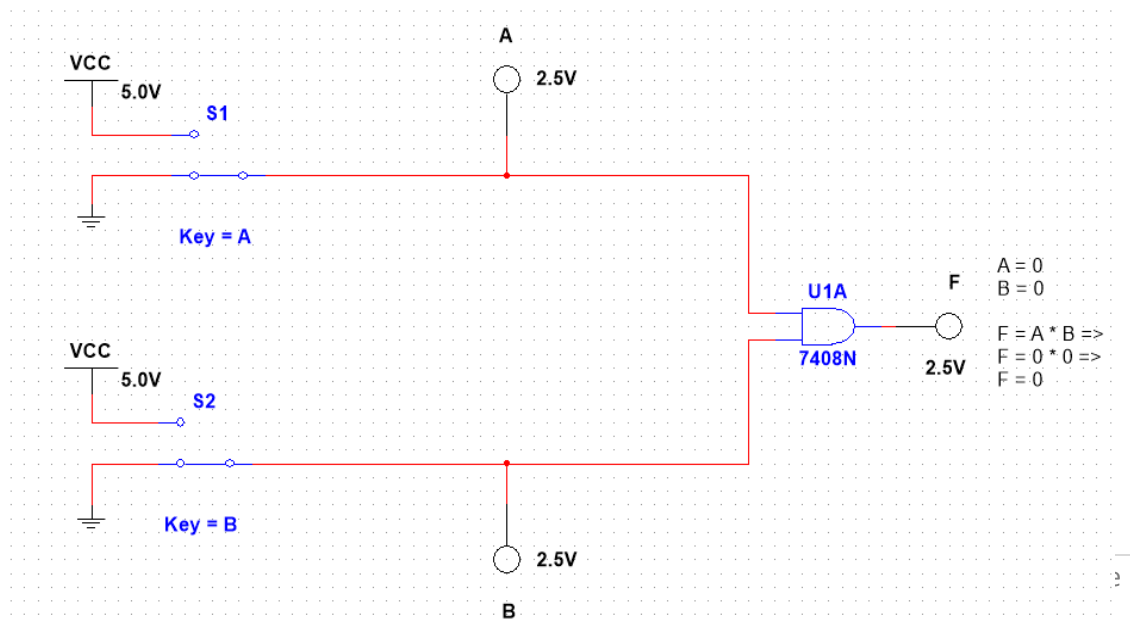
The truth table of the AND gate with inputs A, B and output F is:

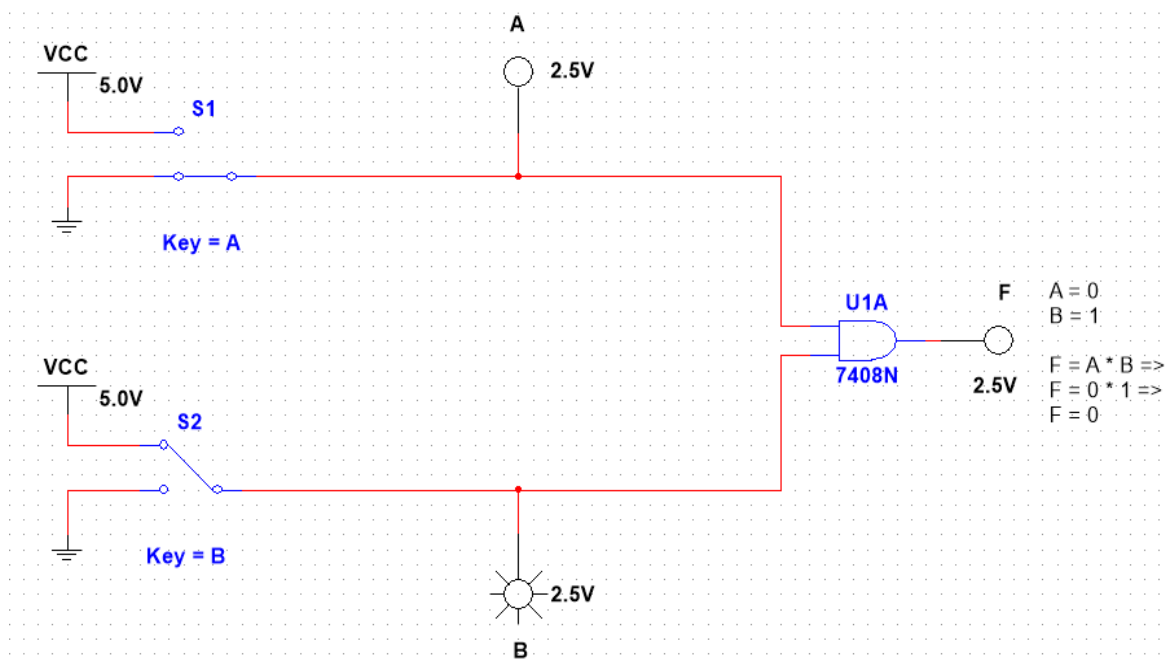
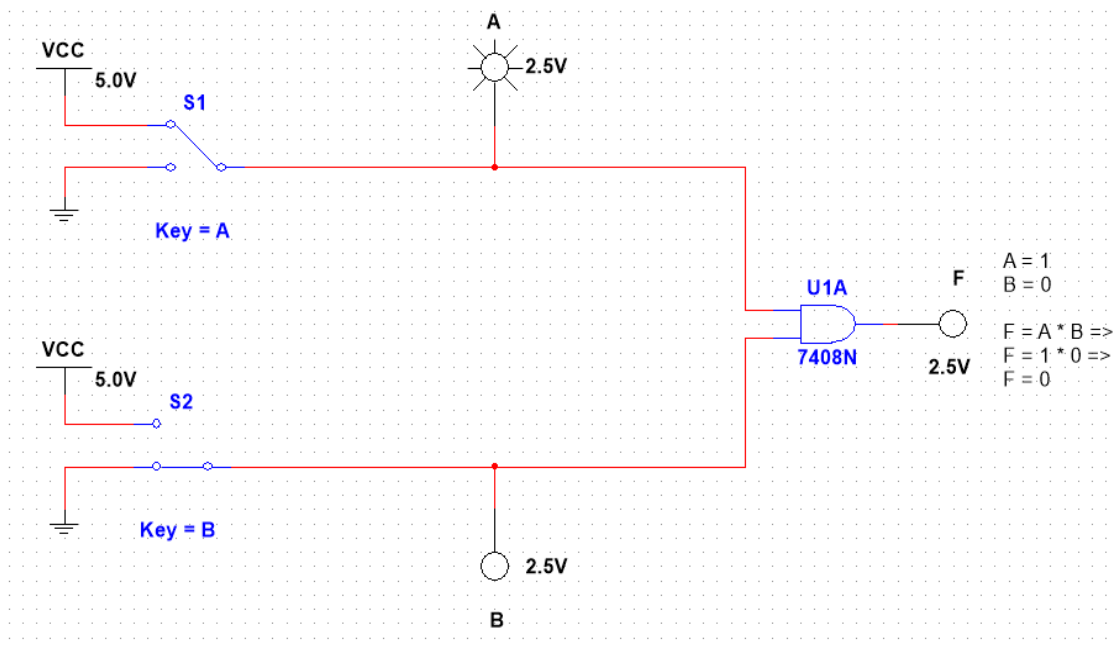
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1

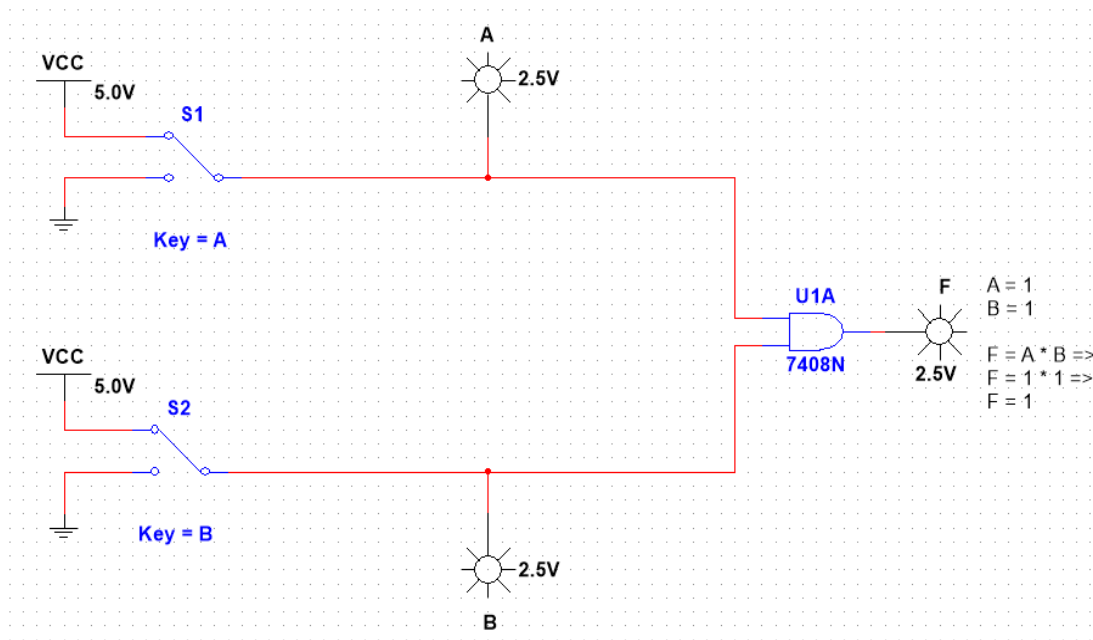
From the truth table, the logical equation for the AND gate is derived:

$$F = A * B$$

ADVERTISEMENT







IMPLEMENTATION

The voltages of the VCC grounds and sources correspond to logic states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp "A" of 2.5 Volt voltage which is also connected to the circuit lights up, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp does not light up. After the output of the logic gate is connected another lamp " F " also of voltage 2.5 Volt , which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up.

According to the above information and specifically for the logic gate " AND " we have:

- i) Lamp A: closed / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp F : closed
- (ii) Lamp A: open / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp F: closed Lamp F : closed

- iii) Lamp A: closed / Lamp B: open => => Lamp A: closed / Lamp B: open
Lamp F : closed
- (iv) Lamp A: open / Lamp B: open => F Lamp F : open

EXERCISE 1.2.2.2 GATES OR , NAND , NOR , EX-OR

THEORETICAL RESULTS

GATE OR

The truth table of gate OR with inputs A, B and output F is:

A	B	F
0	0	0
1	0	1
0	1	1
1	1	1

From the truth table, the logical equation for the OR gate is derived:

$$F = A + B$$

NAND GATE

The truth table of the AND gate with inputs A, B and output F is:

A	B	F
0	0	1
1	0	1
0	1	1
1	1	0

The logic equation for the NAND gate is derived from the truth table:

$$F = \overline{A * B}$$

GATE NOR

The truth table of the NOR gate with inputs A, B and output F is:

A	B	F
0	0	1
1	0	0
0	1	0
1	1	0

From the truth table the logical equation for the NOR gate is derived:

$$F = \overline{A + B}$$

EX-OR GATE

The truth table of the EX-OR gate with inputs A, B and output F is:

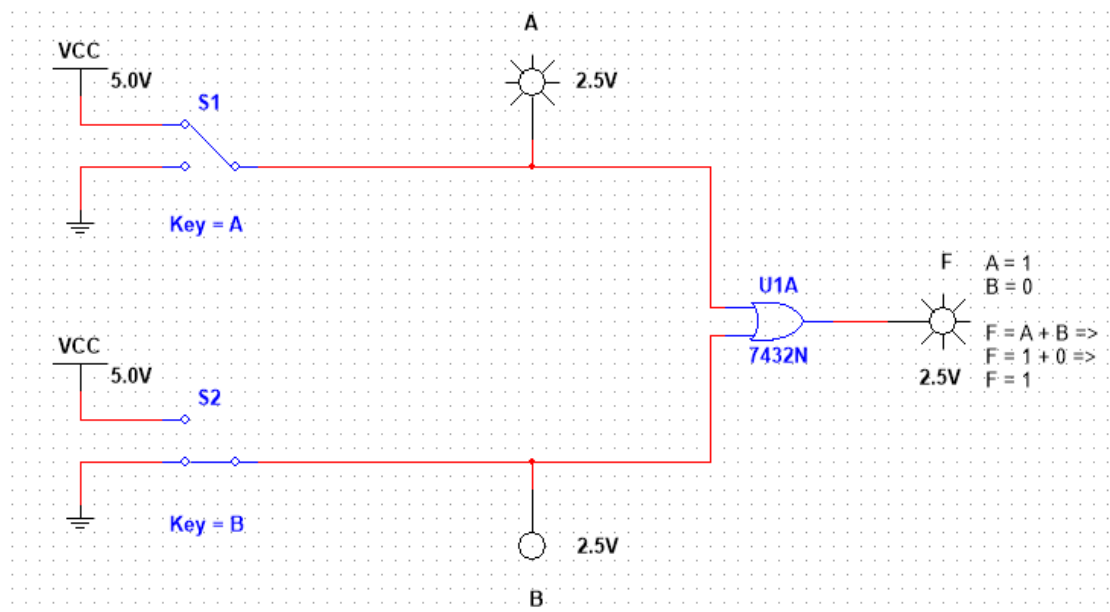
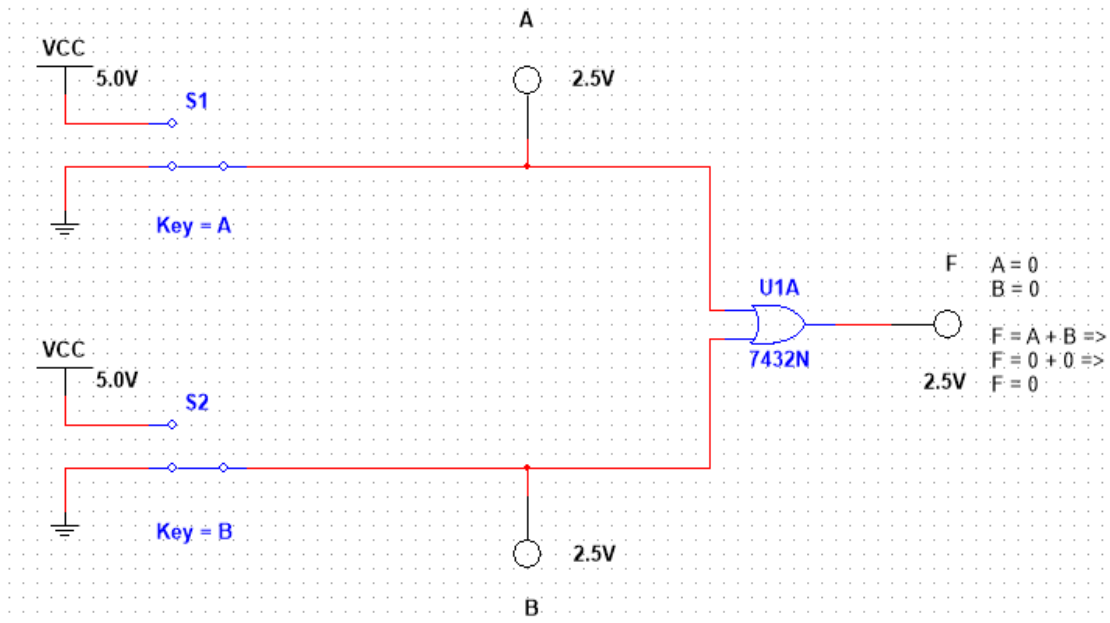
A	B	F
0	0	0
1	0	1
0	1	1
1	1	0

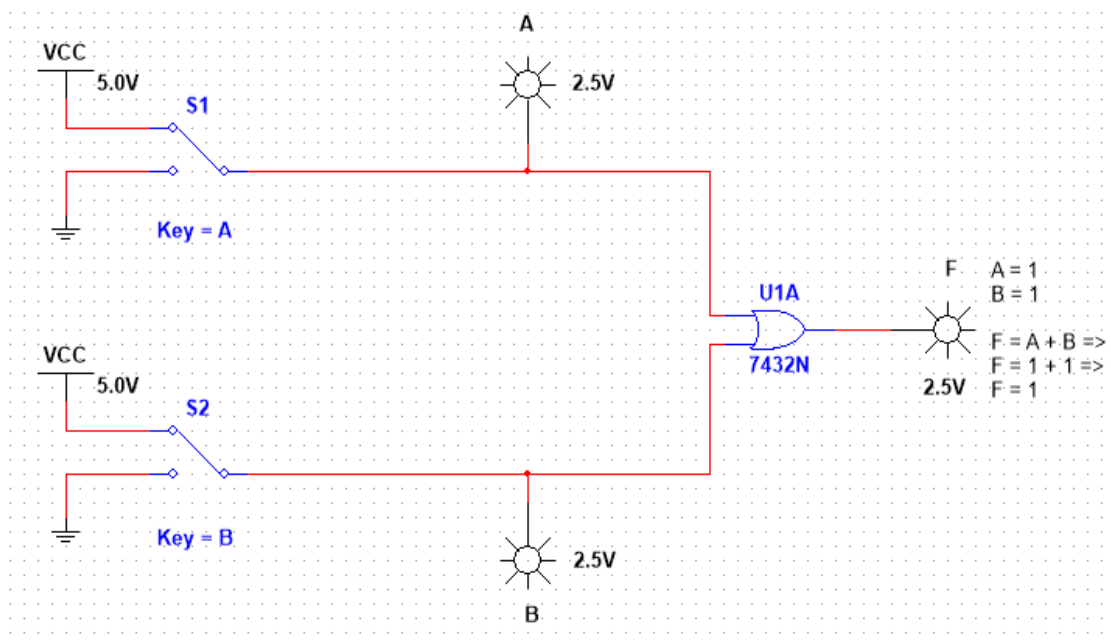
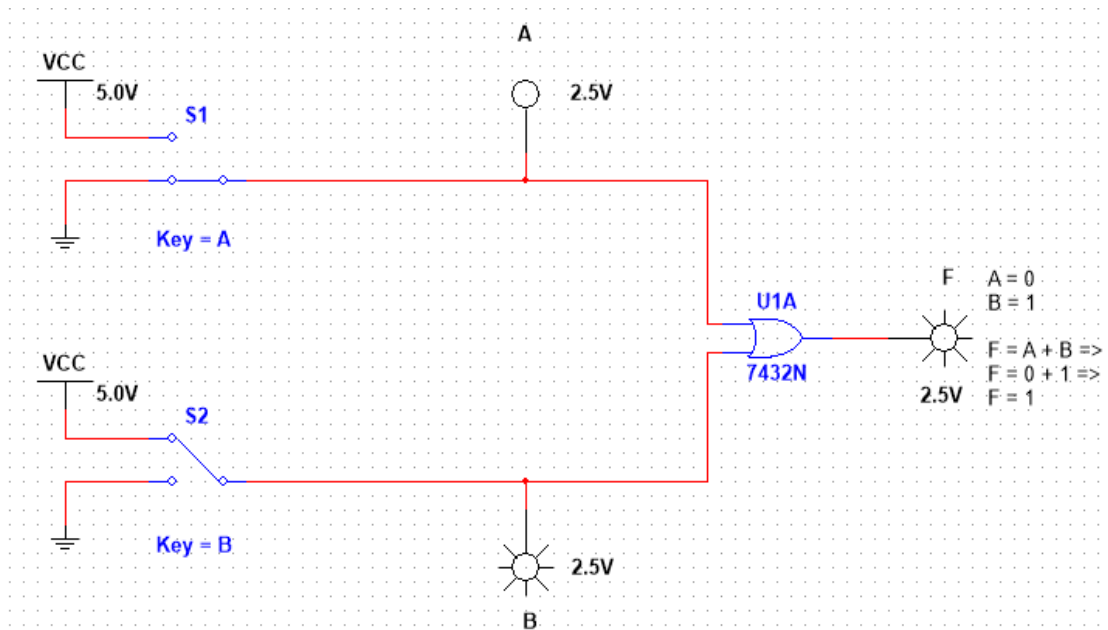
The logic equation for the EX-OR gate is derived from the truth table:

$$F = \overline{A} * B + A * \overline{B}$$

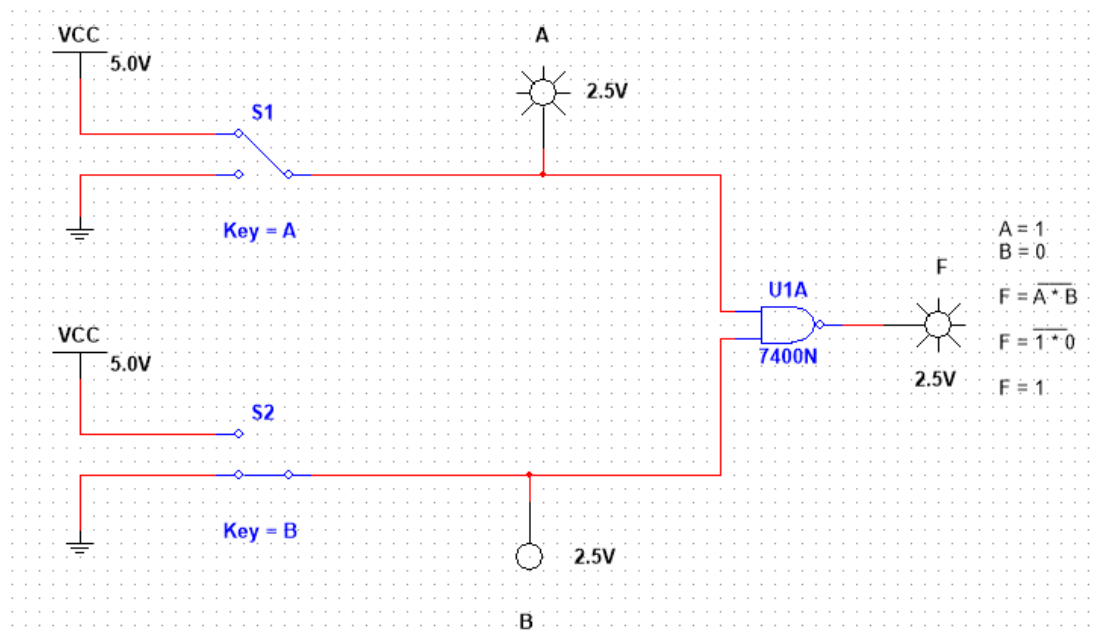
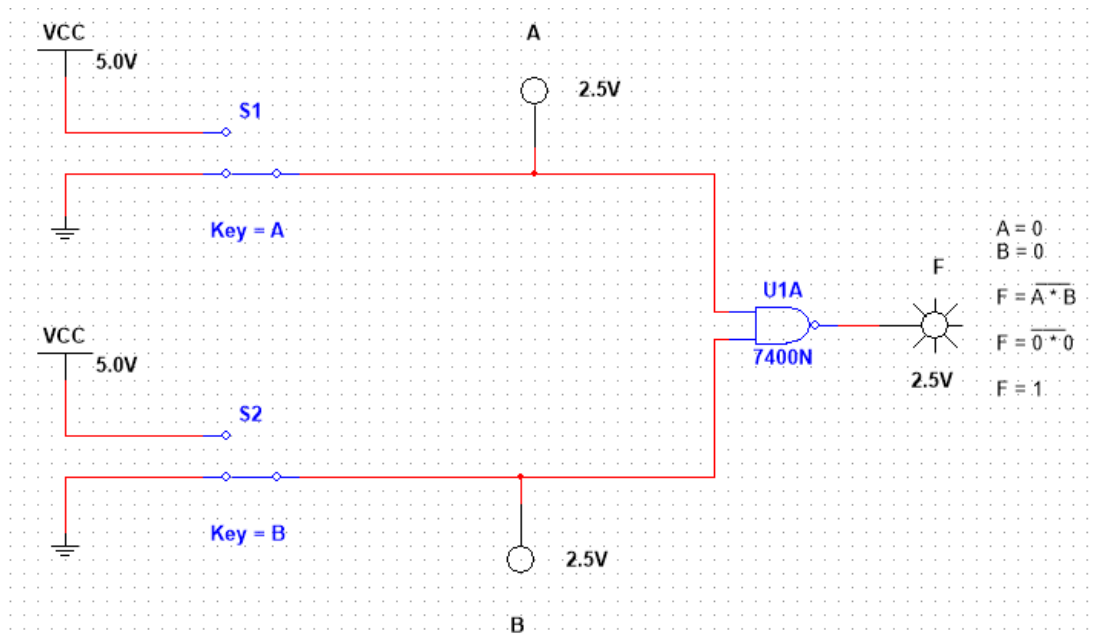
ADVANCE NOTIFICATION

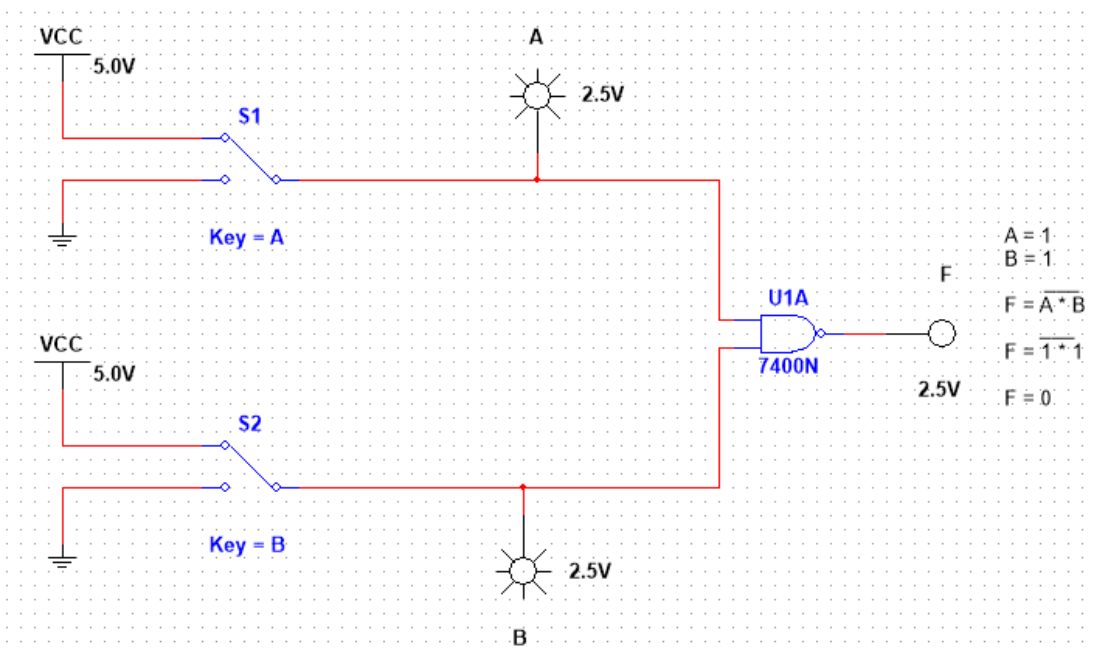
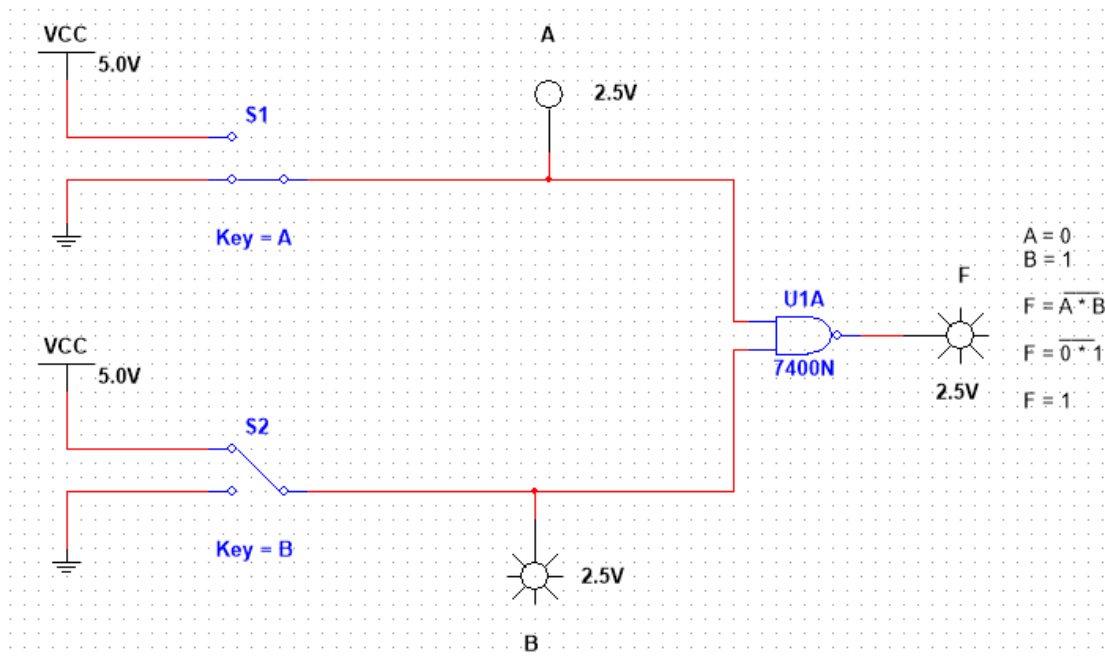
GATE OR



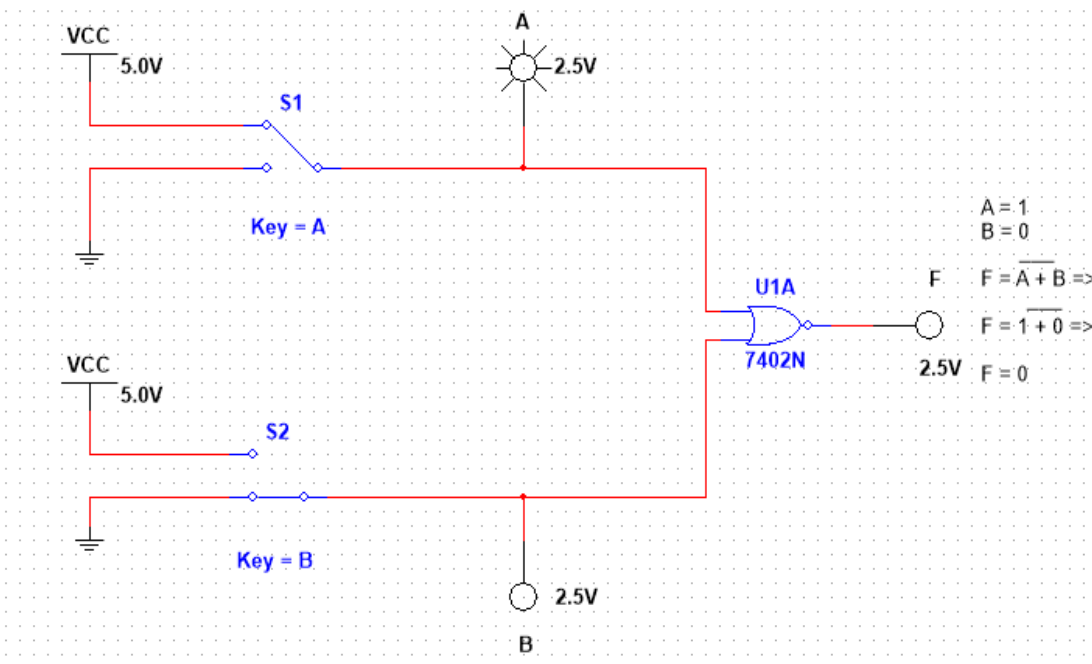
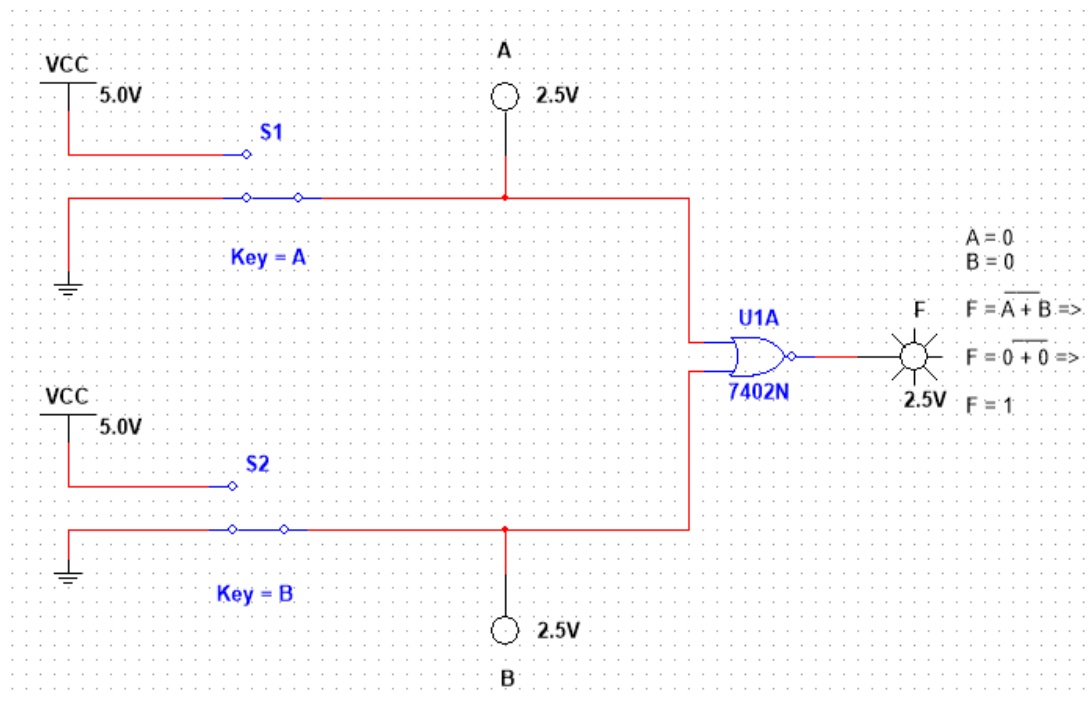


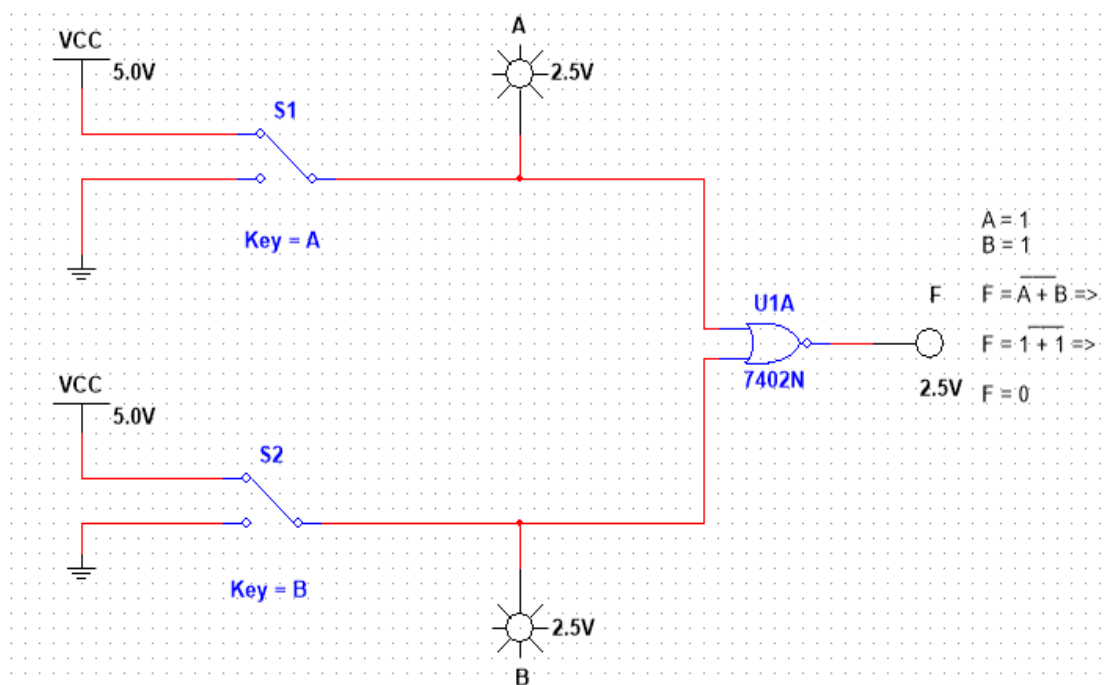
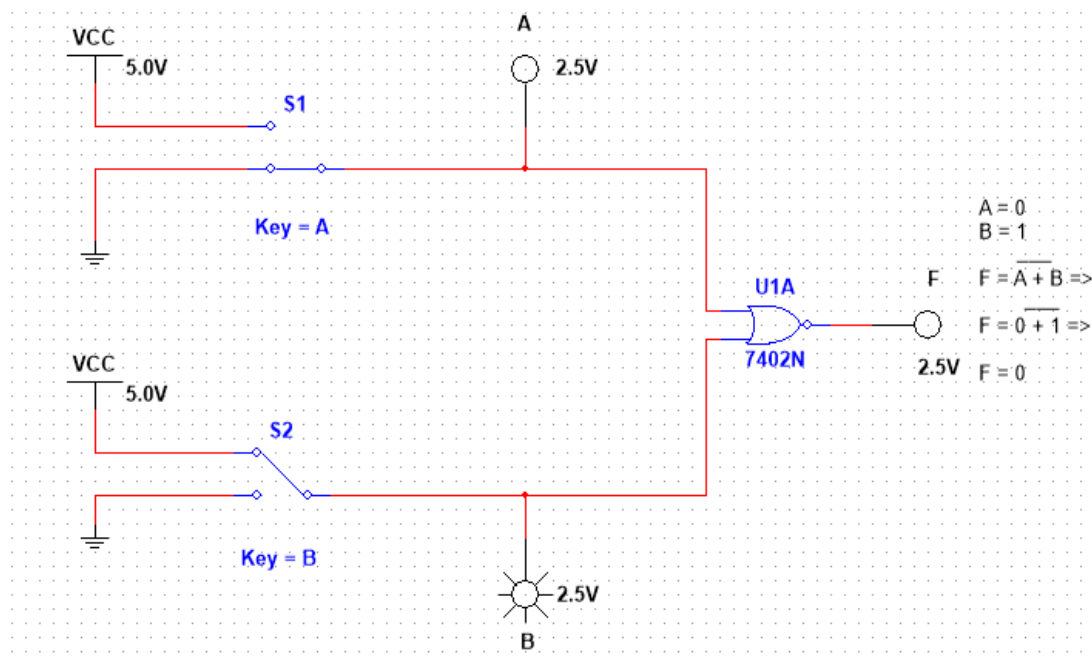
NAND WALL



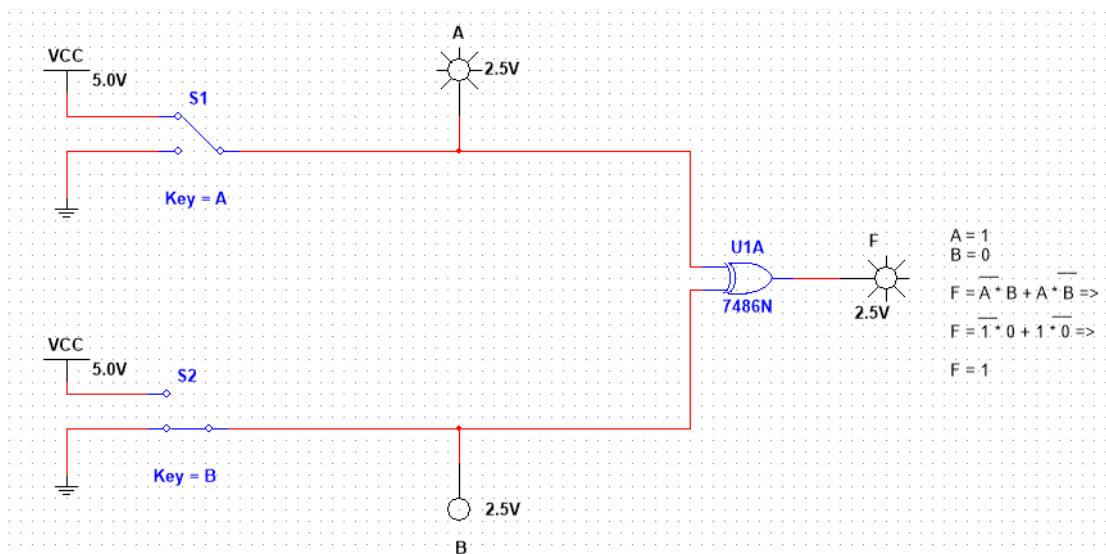
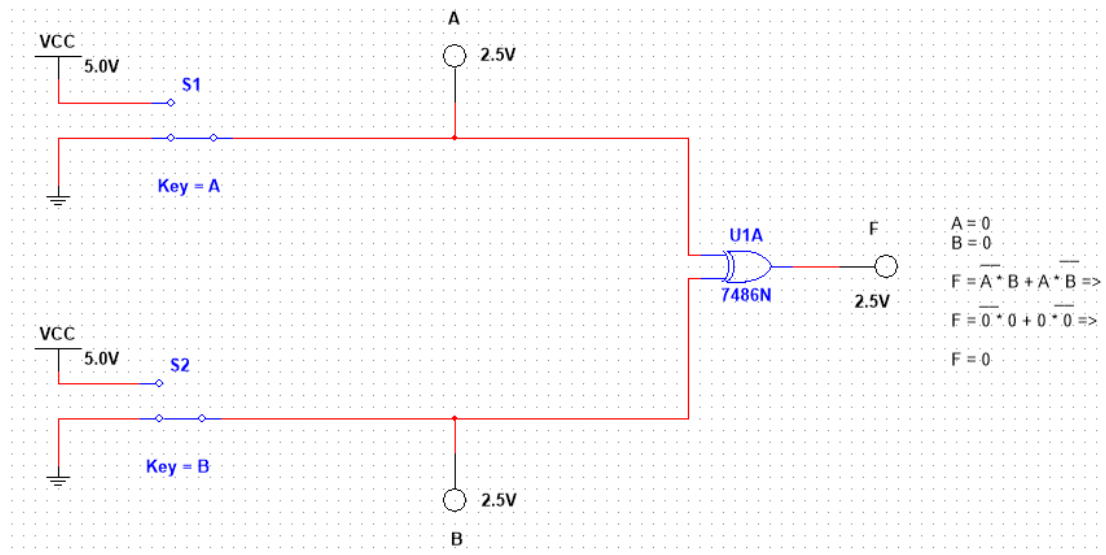


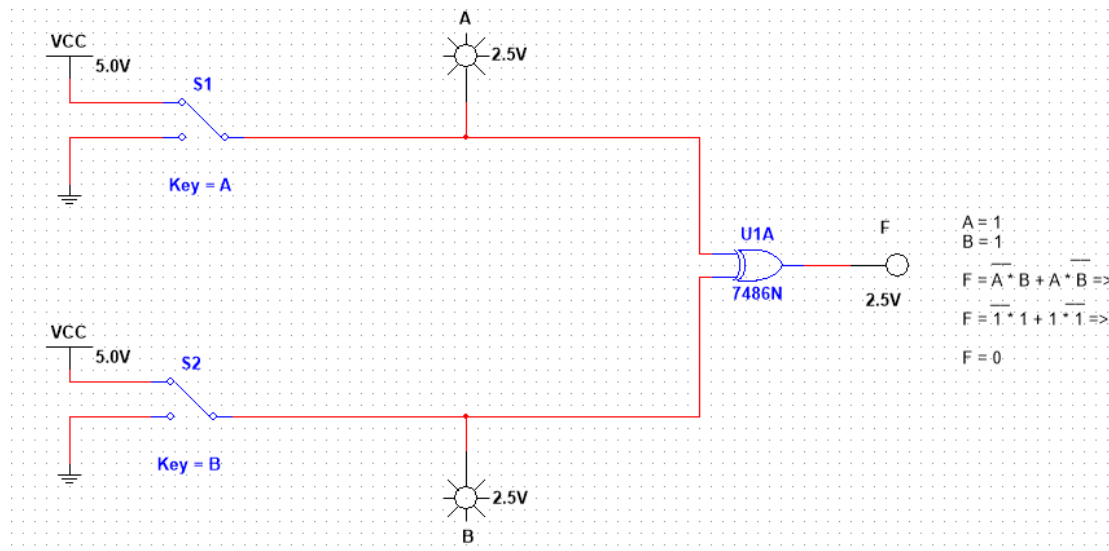
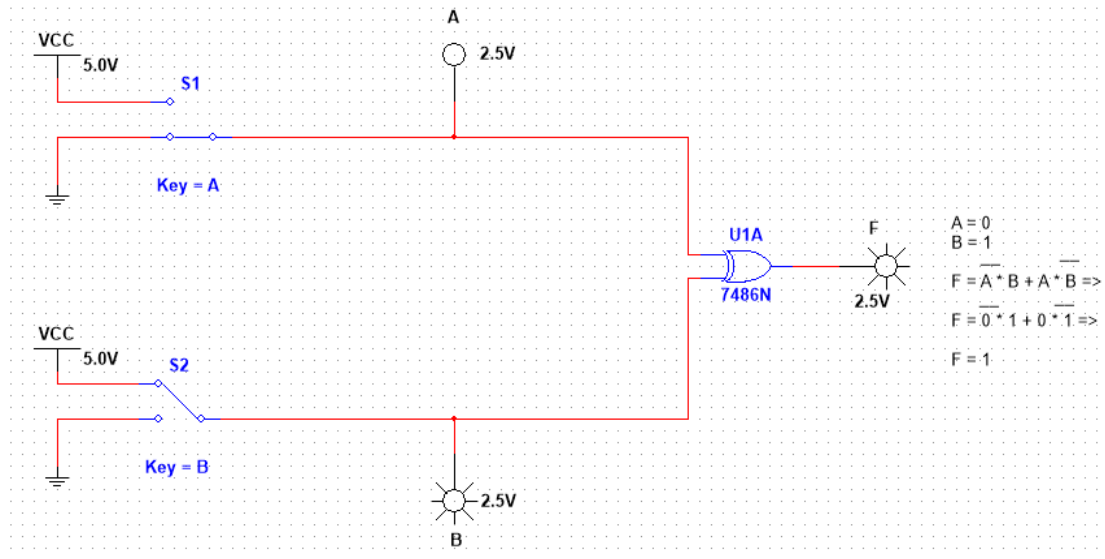
GATE NO





GATE EX-OR





IMPLEMENTATION

The voltages of the VCC grounds and sources correspond to reasonable states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a

lamp "A" of 2.5 Volt voltage which is also connected to the circuit lights up, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp does not light up. After the output of the logic gate is connected another lamp " F " also of voltage 2.5 Volt , which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up.

According to the above information and specifically for the logic gate " OR " we have:

- i) Lamp A: closed / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp F : closed.
- (ii) Lamp A: open / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp F: closed Lamp F : open
- iii) Lamp A: closed / Lamp B: open $\Rightarrow \Rightarrow$ Lamp A: closed / Lamp B: open
Lamp F : open
- (iv) Lamp A: open / Lamp B: open $\Rightarrow \Rightarrow$ Lamp F Lamp F : open

According to the above information and specifically for the logic gate " NAND " we have:

- i) Lamp A: closed / Lamp B : closed $\Rightarrow \Rightarrow$ Lamp F : open
- (ii) Lamp A: open / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp A: open / Lamp B: closed
Lamp F : open
- iii) Lamp A: closed / Lamp B: open $\Rightarrow \Rightarrow$ Lamp A: closed / Lamp B: open
Lamp F : open
- (iv) Lamp A: open / Lamp B: open $\Rightarrow \Rightarrow$ Lamp F : closed

According to the above information and specifically for the logic gate " NOR " we have:

- i) Lamp A: closed / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp F : open
- (ii) Lamp A: open / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp A: open / Lamp B: closed
Lamp F : closed
- (iii) Lamp A: closed / Lamp B: open $\Rightarrow \Rightarrow$ Lamp A: closed / Lamp B: open
Lamp F : closed
- (iv) Lamp A: open / Lamp B: open $\Rightarrow \Rightarrow$ F Lamp F : closed

According to the above information and specifically for the logic gateway " EX-OR " we have:

- i) Lamp A: closed / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp F : closed
- (ii) Lamp A: open / Lamp B: closed $\Rightarrow \Rightarrow$ Lamp A: open / Lamp B: closed
Lamp F : open
- iii) Lamp A: closed / Lamp B: open $\Rightarrow \Rightarrow$ Lamp F Lamp F : open

(iv) Lamp A: open / Lamp B: open => => Lamp F
Lamp F : closed

EXERCISE 1.2.3 X-NOR GATE

THEORETICAL RESULTS

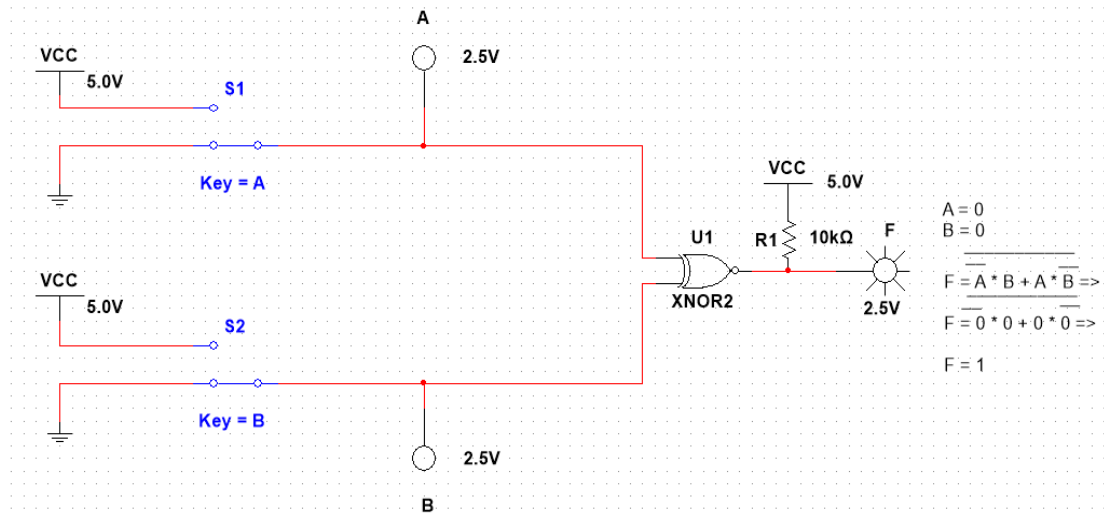
The truth table of the EX-NOR gate with inputs A, B and output F is:

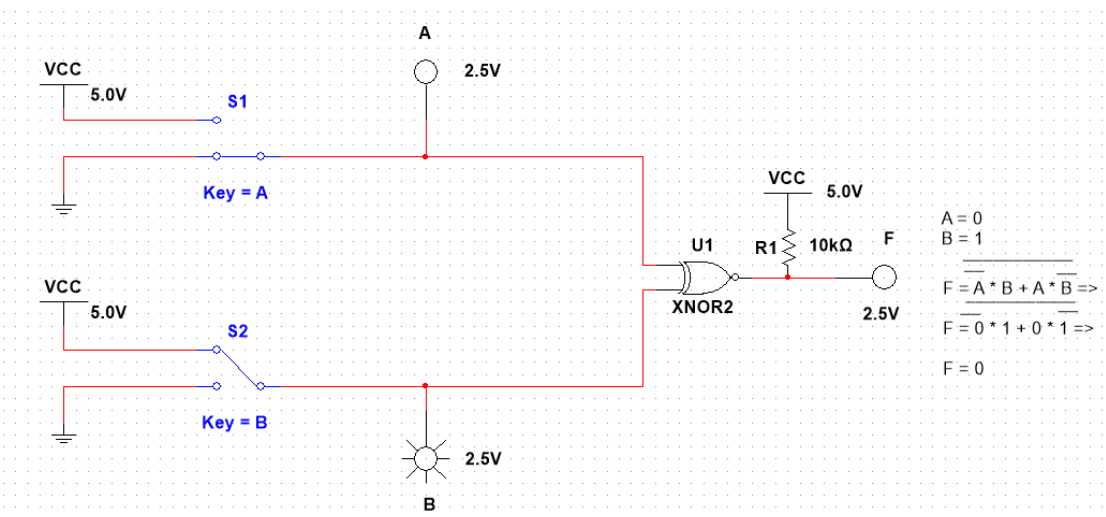
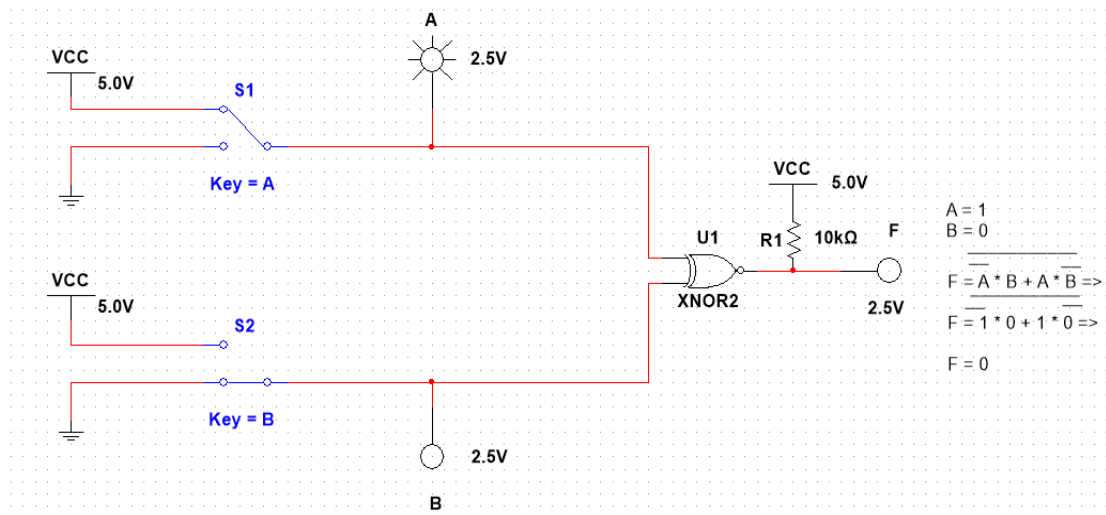
A	B	F
0	0	1
1	0	0
0	1	0
1	1	1

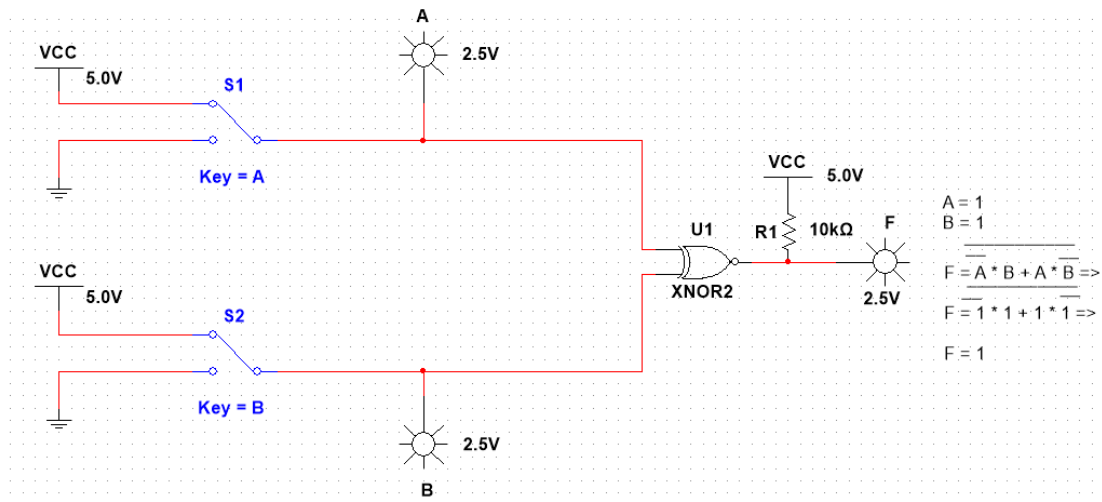
From the truth table the logical equation for the AND gate is derived:

$$F = \overline{A} * B + A * \overline{B}$$

ADVANCE NOTIFICATION







IMPLEMENTATION

The voltages of the VCC grounds and sources correspond to reasonable states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when the switch (S1, S2) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp "A" of 2.5 Volt voltage which is also connected to the circuit lights up, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp does not light up. After the output of the logic gate is connected another lamp " F " also of voltage 2.5 Volt , which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up.

According to the above information and specifically for the logic gate " EX-NOR " we have:

- i) Lamp A: closed/Lamp B: closed => => Lamp F : open
- (ii) Lamp A: open / Lamp B: closed => => Lamp F: closed Lamp F : closed
- (iii) Lamp A: closed / Lamp B: open => => Lamp F Lamp F : closed
- (iv) Lamp A: open / Lamp B: open => F Lamp F : open

EXERCISE 1.2.4 GATE REVERSER NOT

THEORETICAL RESULTS

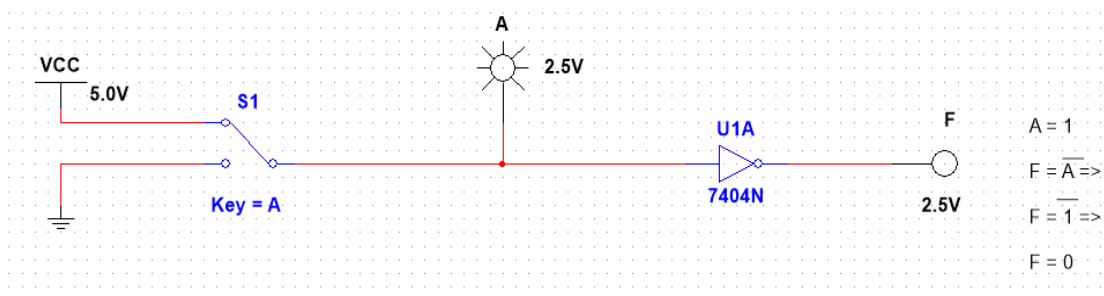
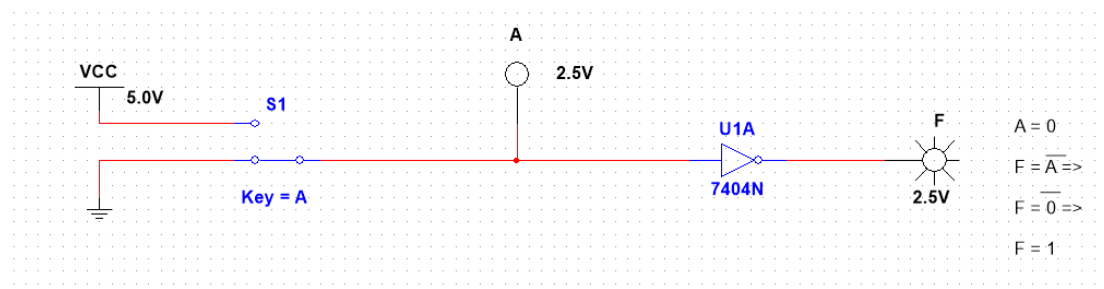
The truth table of the NOT gate with input A and output F is:

A	F
1	0
0	1

From the truth table, the logic equation for the NOT gate is derived:

$$F = \overline{A}$$

PROVISION



IMPLEMENTATION

The voltages of the VCC grounds and sources correspond to reasonable states. In general, a source with a voltage of 0 to 0.5 Volt corresponds to the logical state "0", while a source with a voltage of 2.7 to 5 Volt corresponds to the logical state "1". More specifically, grounding with 0 Volt corresponds to "0" and VCC source with 5 Volt corresponds to "1". So when a switch (S1) is connected to the VCC source then the logic gate connected to the circuit gets an input value of logic "1" and a lamp "A" of 2.5 Volt voltage which is also connected to the circuit lights up, whereas if the switch is connected to the ground then it gets an input value of "0" and the lamp does not light up. After the output of the logic gate is connected another lamp " F " also of voltage 2.5 Volt , which depending on the type of logic gate and the potential connected to the two inputs, lights up or does not light up.

According to the above information and specifically for the logic gate " NOT " we have:

i) Lamp A: closed $\Rightarrow \Rightarrow$ Lamp F : open.

(ii) Lamp A: open $\Rightarrow \Rightarrow$ Lamp F: open Lamp F : closed

EXERCISE 1.2.6 PROPAGATION DELAY

THEORETICAL RESULTS

The propagation delay measurements are:

```
t1 plh 386.321 * 10^-6 ns
t2 plh 387.325 * 10^-6 ns
t1 plh 96.581 * 10^-6 ns
t2 plh 96.8125 * 10^-6 ns
t pav 1.004 * 10^-6 ns
```

ADVANCE NOTIFICATION

